

Statistical Modeling

DATA.STAT.740

Lecture 7

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Effect size and statistical significance

Effect Size

Size of the coefficient or transformed coefficients

The magnitude of the change in the dependent variable for a unit change in the predictor.

Statistical Significance

Represented with a **P-value** that indicates **the likelihood that the observed relationship is due to chance, given the assumption of a true null hypothesis.**

Filters out noise, ensuring the effect is not due to chance. Heavily influenced by sample size - a large N will make tiny, irrelevant effects appear significant (e.g. $p < 0.05$)

Example: A new medication might reduce blood pressure by a *statistically significant* amount, but if the reduction is only 0.1 mmHg, it is *clinically useless*.

Remember the approximate confidence intervals?

Based on asymptotic approximation

$$\hat{\beta} \sim \text{Normal}(\beta, F_{\beta}^{-1})$$

For β_j the standard error of the ML estimator is $\widehat{\text{SE}}(\hat{\beta}_j) = \sqrt{[F_{\beta}^{-1}]_{jj}}$

$100(1 - \alpha)\%$ – CI: $\hat{\beta}_j \pm z_{1-\alpha/2} \widehat{\text{SE}}(\hat{\beta}_j)$

Fast and often reported by software

Wald's hypothesis test

Wald's hypothesis test for coefficients are based on the same approximation

$$\hat{\beta} \sim \text{Normal}(\beta, F_{\beta}^{-1})$$

For β_j the standard error of the ML estimator is $\widehat{SE}(\hat{\beta}_j) = \sqrt{[F_{\beta}^{-1}]_{jj}}$

Formulate a hypothesis pair $H_0 : \beta_j = 0$

$$H_a : \beta_j \neq 0$$

Standard normal rv

Now $W_j = \frac{\hat{\beta}_j}{\widehat{SE}(\hat{\beta}_j)}$ and a P-value is calculated as $p = P(|Z| \geq |W_j| \mid H_0)$

observed
test-statistic

Again: fast and often reported by software (including glm)

Wald's hypothesis test (general form)

Wald's hypothesis test for coefficients are based on the same approximation

$$\hat{\beta} \sim \text{Normal}(\beta, F_{\beta}^{-1})$$

We can formulate a more general hypothesis test pair $H_0 : K^T \beta = 0$

$$H_a : K^T \beta \neq 0$$

If the null H_0 is true, then $K^T \hat{\beta} \sim \text{Normal}(0, K^T (X^T \widehat{W} X)^{-1} K)$

and $Q = (K^T \hat{\beta})^T \widehat{\text{Cov}}(K^T \hat{\beta})^{-1} (K^T \hat{\beta}) = \hat{\beta}^T K (X^T \widehat{W} X)^{-1} K^{-1} K^T \hat{\beta} \sim \chi_q^2$
 $\swarrow$
 $q = \text{rank}(K)$

p-value is $p = P(Q \geq Q_{\text{obs}} \mid H_0)$

Model hypothesis testing

Likelihood ratio test is used to compare two **nested models**

In GLM context we often compare the hypothesis

$$H_0 : g(\mu) = X_1\beta_1$$

$$H_1 : g(\mu) = X_1\beta_1 + X_2\beta_2 = [X_1, X_2] \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$$

LRT is based on the residual deviance

Residual deviance (or just deviance)

Asymptotically equivalent to Pearson

Residual deviance $D(y, \hat{\mu}) = 2 [l(y, y) - l(y, \hat{\mu})]$

$$\text{where } l(y, \mu) = \sum_{i=1}^N \frac{y_i \theta(\mu_i) - b(\theta(\mu_i))}{a(\phi)} + c(y_i, \phi)$$

$l(y, y)$ is called **the saturated model** - every observation gets its own parameter and the fitted mean equal the observed outcome

The best possible log-likelihood any model could achieve on the data

Likelihood ratio test

$$H_0 : g(\mu) = X_1\beta_1$$

$$H_1 : g(\mu) = X_1\beta_1 + X_2\beta_2 = [X_1, X_2] \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$$

Likelihood ratio test statistic

$$\begin{aligned} \Lambda &= -2 \log \left(\frac{\max_{\beta_1} L_{H_0}(\beta_1 \mid y)}{\max_{\beta_1, \beta_2} L_{H_1}(\beta_1, \beta_2 \mid y)} \right) \\ &= -2 \left(l_{H_0}(\hat{\beta}_1 \mid y) - l_{H_1}(\hat{\beta}_1, \hat{\beta}_2 \mid y) \right) \\ &= 2 \left(l(y, y) - l_{H_0}(\hat{\beta}_1 \mid y) \right) - 2 \left(l(y, y) - l_{H_1}(\hat{\beta}_1, \hat{\beta}_2 \mid y) \right) \\ &= D_{H_0} - D_{H_1} \sim \chi^2_{(q)} \end{aligned}$$

$\uparrow$
 $q = \text{rank}(X_2)$

Normal linear model

$$H_0 : g(\mu) = X_1 \beta_1 \quad \begin{matrix} \nearrow \\ N \times p_1 \end{matrix}$$

$$H_1 : g(\mu) = X_1 \beta_1 + X_2 \beta_2 = [X_1, X_2] \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \quad \begin{matrix} \nearrow \\ N \times p_2 \end{matrix}$$

Normal linear model is an exception to likelihood ratio testing in GLM

There is an exact distribution

$$\Lambda = N \log \left(\frac{y^T (I - P_{H_0}) y}{y^T (I - P_{H_1}) y} \right) \sim F_{p_2, N - p_1 - p_2}$$

$$P_{H_0} = X_1 (X_1^T X_1)^{-1} X_1^T$$

$$P_{H_1} = X (X^T X)^{-1} X^T$$

$$X = [X_1, X_2]$$

Likelihood-based goodness-of-fit numbers

Maximum log-likelihood $l(\hat{\beta} \mid y)$

Number of estimated parameters

AIC - Akaike information criterion $AIC = 2k - 2l(\hat{\beta} \mid y)$

Can be used to compare non-nested models

Penalizes model complexity

Lower number is better

BIC - Bayesian information criterion $BIC = \log(N)k - 2l(\hat{\beta} \mid y)$

Can be used to compare non-nested models

Penalizes model complexity (more than AIC when $N > 7$)

Lower number is better

Other goodness-of-fit numbers

In linear regression we used R^2 and adjusted R^2_{adj} to quantify how much of the observation variation a model explains

We can use them for normal linear model with canonical link

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_{i=1}^N (y_i - x_i^T \hat{\beta})^2}{\sum_{i=1}^N (y_i - \frac{1}{N} \sum_{i=1}^N y_i)^2}$$
$$R^2_{\text{adj}} = 1 - \frac{N-1}{N-p-1} (1 - R^2)$$

We can't use the same numbers in GLMs but instead we can pseudo R^2 numbers

Pseudo R^2 numbers

We can't use R squared or adjusted R squared numbers in GLMs, because
Variance depends on the mean,
Outcomes may be discrete (binary, counts),

The log-likelihood plays the role that sum of squared errors plays in linear models

McFadden's pseudo R squared $R_{\text{McF}}^2 = 1 - \frac{l(\hat{\beta})}{l(\hat{\beta}_0)}$
MLE of intercept-only model

Deviance-based R squared $R_{\text{dev}}^2 = 1 - \frac{D(\hat{\beta})}{D(\hat{\beta}_0)}$
deviance of intercept-only model

Example

Dataset: malaria.txt

A random sample of 100 children aged 3-15 (x_1) years from a village in Ghana. The children were followed for a period of 8 months. At the beginning of the study, values x_2 of a particular antibody were assessed. Based on observations during the study period, the children were categorized into two groups: individuals with and without symptoms of malaria y_i .

Example

Dataset: tetrahymena.txt

Consider the research problem where there was interest to determine whether glucose had an effect on single-cell Tetrahymena cell size when cells were grown in cell culture on the growth medium. The cell size was measured by the (average) diameter of cells y_i . In research, explanatory variables were the concentration of Tetrahymena cells x_1 and glucose x_2 which was coded

$$x_2 = \begin{cases} \text{yes} & \text{Glucose was added to growth medium} \\ \text{no} & \text{Glucose was not added to growth medium} \end{cases}$$

Example

Dataset: species.txt

The response is a count of the number of plant species on plots that have different biomass (a continuous explanatory variable x_1) and different soil pH (a categorical variable x_2 with three levels: high=1, mid=2, and low=3).

Example

Dataset: canoladiesel.txt

In a biodiesel study, methyl ester was produced from waste canola oil. Through experiments, the effects of process time (x_1), temperature (x_2) and the level of Methanol/Oil weight ratio (x_3) to the yield of methyl ester was observed.

End of Lecture 7